

Performance–cost balance in large language model multi-agent systems for medical examination question answering: a systematic review

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Abstract

Background. Large language model (LLM) multi-agent systems (MAS) are increasingly proposed for medical question answering. Published frameworks are almost uniformly optimised for accuracy, while the inference cost they incur is rarely measured and almost never traded off explicitly.

Objectives. To determine (i) what single-agent baselines and evaluation slices the literature actually uses, (ii) how accuracy gain and cost vary with the number of agents and backbone model scale, (iii) which collaboration mechanisms buy accuracy cheaply, and (iv) how gains vary with item difficulty.

Methods. We searched PubMed and arXiv (2022-01-01 to 2026-09-23), supplemented by Google Scholar, ACL Anthology and forward citation chasing, for LLM-based multi-agent systems evaluated on text-based medical examination question answering. Peer-reviewed publications were prioritised and the publication status of every record was verified against Semantic Scholar and OpenAlex. The primary outcome is a *pair*: the within-study paired accuracy difference ΔAcc against a matched single-agent baseline, and the cost ratio R . Because comparisons cluster within studies, the study – not the comparison – is the unit of inference; uncertainty is quantified by a study-clustered bootstrap.

Results. 2,288 deduplicated records yielded 343 for manual review and 202 retrievable full texts. Thirty studies contributed 293 extractable comparisons, 280 of them from peer-reviewed sources. Restricted to exam-style medical multiple-choice accuracy, peer-reviewed sources and systems whose multi-agent contribution is separable from fine-tuning, **93 comparisons from 13 studies** were available against the strongest single-agent configuration reported in the same study. The comparison-level median was -1.73 pp (IQR -4.62 to $+0.64$; 69% negative; only 14% exceeded the pre-specified 2 pp threshold), the **study-level median was -0.31 pp** (7 of 13 studies negative), and the study-clustered bootstrap 95% CI for the pooled median was $[-2.70, +0.36]$, **which includes zero**. The reported benefit fell monotonically with comparator strength: $+8.4$ pp against chain-of-thought with self-consistency, $+7.7$ against zero-shot, $+4.7$ against a same-scale single model, $+1.9$ against the same backbone without retrieval, and -1.7 against the strongest single agent. Typical cost was a median R of $4\times$. Gains concentrated on hard items: against a same-backbone single agent the median effect was -3.1 pp on Basic items, -2.2 on Intermediate and $+1.4$ on Expert.

Conclusions. There is *no reproducible accuracy advantage* for medical multi-agent systems once the comparator is the strongest single agent the same paper already ran; the direction is robust across every sensitivity analysis but the magnitude is not. Reporting of cost is so incomplete that cost had to be derived from architecture in most comparisons. We recommend that future work report a same-backbone strongest baseline, an equal-budget self-consistency control arm, and per-question cost as a matter of course.

Registration. Protocol in `design/plan.md`; *not* prospectively registered (see §4.3).

1 Introduction

Since instruction-tuned LLMs began to approach passing scores on medical licensing examinations, a large family of *multi-agent* systems has been proposed: panels of specialist personas, debating agents, generator–critic pairs, hierarchical coordinators and heterogeneous model ensembles. The rhetorical structure of these papers is remarkably uniform: a new orchestration is introduced, evaluated on MedQA, MedMCQA or the medical subsets of MMLU, and shown to exceed some single-model baseline.

Two things are almost always missing. First, the **cost** of the orchestration. A system that issues seven model calls per question is not comparable with one that issues a single call, yet accuracy is typically reported as if it were. Second, a **fair comparator**. When the baseline is a zero-shot prompt while the proposed system uses chain-of-thought reasoning, several rounds of sampling, retrieval and an aggregation step, the reported “multi-agent gain” confounds collaboration with everything else that was added.

This review therefore does not ask *which framework is best*. It asks where the **balance** lies: what combination of model scale, agent count, mechanism and item difficulty buys a meaningful accuracy gain at an acceptable cost.

Research questions.

RQ1 Which baselines, dataset slices and sampling procedures does the literature use, and how comparable are the resulting evaluation groups?

RQ2 How do ΔAcc , R and CEI vary with the number of agents and backbone scale?

RQ3 Which mechanisms (debate, role specialisation, retrieval, heterogeneity, verification, aggregation) deliver accuracy cheaply?

RQ4 How do gains vary with item difficulty?

2 Methods

2.1 Protocol and reporting

The review follows PRISMA 2020 for reporting, PRISMA-S for the search, and SWiM for synthesis without meta-analysis. The protocol is version-controlled in `design/plan.md`. Deviations from it are listed in §4.3.

2.2 Eligibility criteria

Included: studies evaluating an LLM-driven system in which at least two agents exchange information before producing an answer (or an internal ablation of such a system), reporting accuracy on a text-based medical examination-style multiple-choice benchmark, first released on or after 2022-01-01, in English or Chinese. **Publication type:** peer-reviewed journal articles and papers accepted at peer-reviewed conferences (in this field the conference *is* the venue of record) were prioritised. Preprints were retained in the register but enter the primary analysis only when the evidence they contain is not available from a peer-reviewed source; every extracted row carries its publication status. **Excluded:** primarily multimodal inputs; primarily open-ended generation without an objective key; no extractable numerical result; and systems for which the multi-agent contribution cannot be separated from retrieval, tools or extra sampling *and* which report no ablation.

2.3 Information sources and search

The search was run in two layers. The **horizontal layer** covered all four research questions plus the cross-cutting cost dimension; the **vertical layer** deepened each dimension with question-specific terminology, giving a matrix of 35 queries (RQ1 evaluation and baselines, 6; RQ2 agent count and model scale, 7; RQ3 mechanisms, 10; RQ4 difficulty, 6; cost and efficiency, 6). The matrix was executed against three bibliographic APIs and, in addition, against Google Scholar through a browser, because Scholar indexes conference proceedings (ACL Anthology, AAAI, NeurIPS, ICASSP, ACM) that the other sources cover unevenly.

Sources and yields: PubMed (E-utilities, 752 records), OpenAlex (matrix queries, 461+ records), arXiv (facet queries, supplementary probes and the full matrix, 2,592 records), Google Scholar (69 harvested records with venue and citation counts), and forward citation chasing through the Semantic Scholar graph on six seed papers (MedAgents, MDAgents, MedAgent-Board, *More agents is all you need*, multi-agent debate, Medprompt), which returned 722 citing works. All searched to 2026-09-23.

Google Scholar proved decisive: it surfaced 34 records absent from the API corpus, including the most-cited works in the field and an entire cost-aware multi-agent literature (token-distribution benchmarking, budgeted routing, cost-aware agent pruning) that the medical-facing queries had missed. Open-access full texts of peer-reviewed articles were obtained through PubMed Central and through open-access links resolved via OpenAlex. Publication status of every arXiv record was verified against the Semantic Scholar batch API, which identified 28 of 101 retrieved arXiv items as papers of record at NeurIPS, ICLR, ACL, EMNLP, AAAI, KDD, *npj Digital Medicine*, *PLOS Digital Health*, JAMIA and IEEE venues. The search combined three facets — multi-agent terminology, LLM terminology and medical examination QA terminology — with OR within facets and AND between them. Variables that are reported only in the body of papers (retrieval use, parameter count, cost, agent count) were deliberately *not* placed in the query, since doing so destroys sensitivity.

2.4 Selection

Records were triaged by a pre-specified rule into three tiers, and tier-1 records (all three facets plus a medical term) were reviewed manually. Full texts were retrieved automatically where an open-access PDF existed.

2.5 Outcomes and derived quantities

For each comparison we recorded the multi-agent accuracy Acc_{MAS} , the baseline accuracy Acc_{base} and the comparator type. The primary outcome is the pair

$$\Delta\text{Acc} = \text{Acc}_{\text{MAS}} - \text{Acc}_{\text{base}}, \quad R = \frac{\text{calls per question (MAS)}}{\text{calls per question (baseline)}},$$

with derived quantities $\text{RRB} = \Delta\text{Acc}/(100 - \text{Acc}_{\text{base}})$, $\text{CEI} = \Delta\text{Acc}/\log_2 R$ and $\text{ICAR} = (R - 1)/\Delta\text{Acc}$. Where cost was not reported — the usual case — R was derived from the architecture description (agents $\times$ rounds, plus any aggregation call) and flagged as derived. An interpretation threshold of $\text{MID} = 2.0$ pp was set a priori.

2.6 Contrast typology

Comparisons were classified as **Type A** (within-paper, MAS versus a matched single-agent baseline), **Type B** (within-paper ablation varying exactly one factor) or **Type C** (across papers within the same evaluation cell). Only Type B evidence is treated as capable of supporting a claim about moderation; Type C differences are reported descriptively.

2.7 Synthesis

Paired variances were recoverable in almost no study, so no meta-analysis was attempted. Results are summarised as medians with interquartile ranges, grouped by comparator type, mechanism, agent count and difficulty, following SWiM.

3 Results

3.1 Search and selection

The pre-specified triage rule was applied to **two** sources: 1,536 arXiv records and 752 PubMed records, 2,288 after deduplication, of which **343** were assigned to manual review. A further 1,129 OpenAlex records were retrieved but *never entered the triage rule*, because the API returns titles without retrievable abstracts; they could only be screened post hoc at title level, where 0 of 1,070 unscreened records passed a three-facet screen and the 3 that passed a two-facet screen were clearly ineligible. Search sensitivity is therefore bounded below rather than established (§4.3). Retrieval yielded **84 peer-reviewed articles as PDFs** (56 open-access journal articles plus 28 arXiv-archived papers of record at peer-reviewed venues), **67 further peer-reviewed articles as full text** where the publisher blocked programmatic PDF access, and 73 preprints; 50 subscription articles could not be retrieved and are listed for manual retrieval, together with 6 failed downloads, in `documents.md`.

Thirty studies yielded **293 extractable comparisons**, of which **280 come from peer-reviewed sources**. Comparisons were stratified by evaluation tier: 158 *primary* (exam-style medical multiple-choice accuracy), 106 *secondary* (biomedical literature and educational QA), 21 *evidence map* (non-multiple-choice medical outcomes) and 9 *background* (general-domain). The primary analysis uses the primary tier only, from peer-reviewed sources, excluding systems whose multi-agent contribution cannot be separated from fine-tuning on the evaluation benchmarks' training splits.

3.2 RQ1: the comparator determines the conclusion

Table 2 and Figure 2 give the central result of this review. Grouped by the comparator that the primary study chose, the median reported gain falls monotonically as the comparator gets stronger: +8.4 pp against chain-of-thought with self-consistency, +7.7 pp against a plain zero-shot prompt, +5.5 pp against a single agent on the same backbone, +4.7 pp against a single model of the same scale, +1.9 pp against the same backbone without retrieval, and **−1.73** pp against the strongest single agent available in the same study.

The unit of inference is the study, not the comparison. Ninety-three comparisons is not ninety-three independent observations: they come from 13 studies, one of which contributes 36 of them. We therefore report both levels (Figure 1). At the comparison level the median is **−1.73** pp, 69% are negative and only 14% exceed the 2 pp threshold. At the study level the median of the 13 study medians is **−0.31** pp with 7 of 13 studies negative, and a study-clustered bootstrap gives a 95% CI for the pooled median of $[-2.70, +0.36]$. **That interval contains zero.** The defensible claim is therefore the *absence of a reproducible gain*, not a reliable deficit of about two points; reporting a comparison-level *p*-value here would be spurious precision.

Table 1 shows that the direction survives every sensitivity analysis we pre-specified, while the magnitude does not. Leaving out the single largest contributor moves the comparison-level median from **−1.73** to **−0.60** pp and the negative fraction from 69% to 60%; across all thirteen leave-one-study-out refits the median ranges from **−2.12** to **−0.60** pp and the negative fraction from 60% to 72%. No refit produces a positive median.

Table 1: Sensitivity analyses. k = studies; CI is a study-clustered bootstrap (4,000 resamples) for the pooled median. Every specification places the median at or below zero and every interval includes zero.

Specification	n	k	Comparison median	Study median	95% CI
Primary analysis	93	13	−1.73	−0.31	[−2.70, +0.36]
Excluding the largest study	57	12	−0.60	−0.14	[−2.55, +0.80]
Primary + secondary evaluation tiers	190	13	−1.65	−0.39	[−2.50, +0.41]
Without the fine-tuning exclusion	98	14	−1.55	−0.14	[−2.60, +0.80]

The decisive single study is MedicalAgentsBench [Tang et al., 2026], which evaluates four multi-agent methods and five single-agent prompting configurations across nine medical datasets and four backbones on a difficulty-curated hard subset. Of its 108 comparisons against the strongest single-agent configuration in the same cell, 78 (72%) are negative; restricted to the primary tier the median is −3.30 pp with 83% negative, and restricted further to the two discussion-based systems it is −4.15 pp with 92% negative. MedAgentBoard [Zhu et al., 2026] reaches the same conclusion on a separate corpus: on MedQA, chain-of-thought with self-consistency scores $89.90 \pm 2.43\%$ while MedAgents, ColaCare, MDAgents and ReConcile score 85.25, 84.65, 78.80 and 78.00 — every multi-agent system below the best single agent. In that same paper MedAgents is +7.75 pp when the comparator is instead zero-shot prompting, which demonstrates the comparator effect *within a single study*. Its authors state the conclusion directly: multi-agent collaboration “does not consistently outperform advanced single LLMs”.

Table 2: Reported ΔAcc by comparator type. Peer-reviewed studies only, restricted to the primary evaluation tier (exam-style medical multiple-choice accuracy), excluding systems whose multi-agent contribution cannot be separated from fine-tuning on the evaluation benchmarks’ training splits. Quartiles use the exclusive definition (`statistics.quantiles`).

Comparator used by the primary study	n	Median ΔAcc (pp)	Negative
Chain-of-thought + self-consistency	4	+8.38	0%
Zero-shot prompt	10	+7.72	0%
Single agent, same backbone	8	+5.53	0%
Single model of the <i>same scale</i>	6	+4.67	0%
Same backbone, no retrieval	16	+1.91	38%
Strongest single agent available	93	−1.73	69%
<i>Aggregation:</i> judge arm vs majority voting	3	+4.00	0%
<i>Aggregation:</i> majority voting vs strongest single	3	−0.71	67%

The largest controlled benchmark agrees. MedAgentsBench [Tang et al., 2026] evaluates four multi-agent methods and five single-agent prompting configurations across nine medical datasets and four backbones, on a difficulty-curated hard subset (862 items). Of its 108 comparisons against the strongest single-agent configuration in the same cell, **78 (72%) are negative**; restricted to the primary evaluation group (MedQA, MedMCQA, the six MMLU medical subjects) the median is −3.30 pp with **83% negative**, and restricted further to the two discussion-based systems (MedAgents, MDAgents) it is −4.15 pp with **92% negative**.

3.3 RQ2: more agents is not better, and a non-interactive control often wins

The clearest within-paper evidence comes from Yun et al. [2026] and Mehrabi et al. [2026], who evaluate six single-agent experts, six-agent interactive baselines and their own systems on the same MedMCQA slice (Table 3).

Two observations follow. First, *interaction per se does not pay*: four of the six interactive

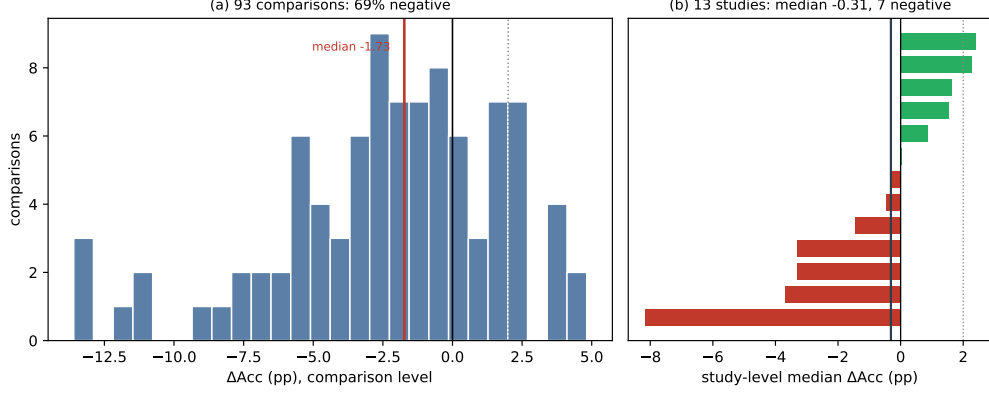


Figure 1: (a) Distribution of the 93 peer-reviewed comparisons against the strongest single-agent configuration reported in the same study; red line = comparison-level median (-1.73 pp), dotted line = the 2 pp threshold. (b) The same evidence aggregated to its unit of inference: one median per study. Seven of thirteen study medians are negative and the median of medians is -0.31 pp.

Table 3: MedMCQA accuracy under a common evaluation protocol and a common six-model pool [Yun et al., 2026, Mehrabi et al., 2026]. Δ is against the best single agent (55.22).

System	Interaction	Agents	Acc. (%)	Δ (pp)
Best single agent (general)	—	1	55.22	—
Self-consistency	none (resampling)	1	55.70	+0.48
Self-MoA	homogeneous mixing	6	55.56	+0.34
Mixture-of-Agents	mixing	6	54.94	-0.28
Self-refine	iterative	6	54.94	-0.28
ReConcile	debate + confidence	6	54.60	-0.62
Multi-agent debate	debate	6	53.05	-2.17
Graph-of-Agents (selected)	graph message pass	3	60.04	+4.82
Relational aggregation	structured passing	6	62.83	+7.61

six-agent baselines score below the best single agent, while non-interactive self-consistency — which requires no communication protocol at all — is slightly positive. Second, *agent count is not the lever*: the best-performing configuration in this pool uses three selected agents and outperforms six-agent systems, consistent with the authors’ own framing.

Orchestrating small models loses to simply using a bigger one. The sharpest statement of the balance problem comes from S-DAG [Authors, 2026], which evaluates six multi-model orchestration methods on MedMCQA using a pool of 7–13B experts, alongside single-model chain-of-thought baselines at several scales. Measured against a *same-scale* single 7B model (72.07%), every orchestration method is positive: median +4.67 pp (range +2.51 to +6.48). Measured against the strongest single model in the same table — Qwen2.5-72B with plain chain-of-thought at 80.44% — *every one of them is negative*: median -3.71 pp (range -5.86 to -1.89), including the paper’s own system at -2.06 pp.

Table 4 makes the practical implication explicit: four to six orchestrated calls to 7B models cost more than one call to a 72B model and deliver less accuracy. Any balance claim that omits the larger single model as a comparator is therefore not a balance claim at all.

Backbone scale otherwise behaved inconsistently. In Tang et al. [2023] the same architecture gained +9.8 pp on GPT-3.5 and +10.7 pp on GPT-4 over a zero-shot baseline, i.e. no attenuation with scale, whereas Su et al. [2025] show a 7B student with self-collaborative correction reaching 71.3% on CMExam, above its own 32B teacher (67.5%). We therefore found no consistent

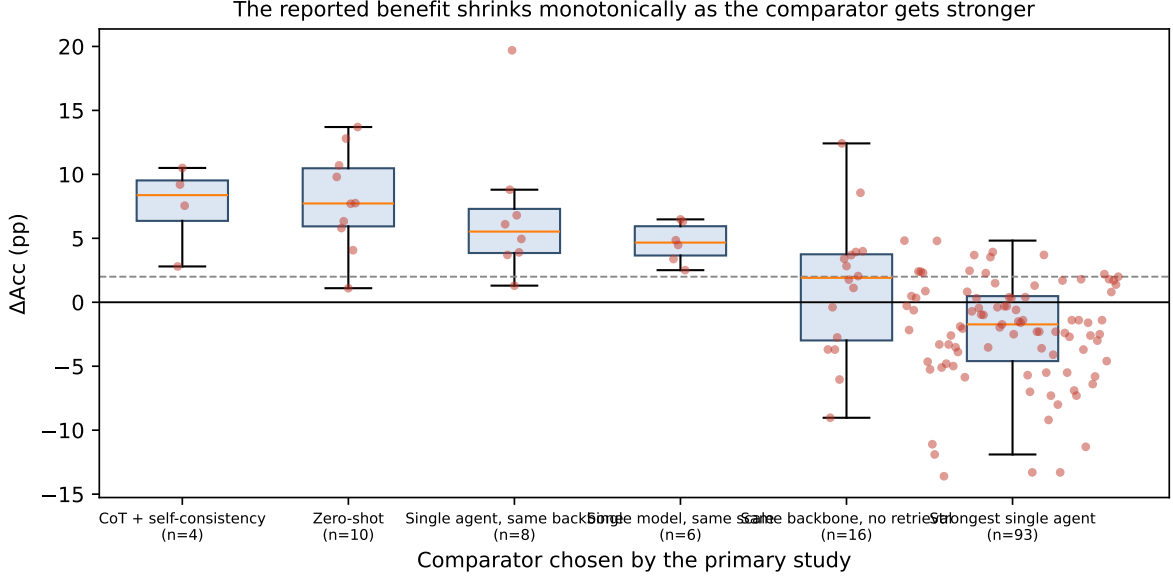


Figure 2: Each point is one extracted comparison; circles are peer-reviewed sources, triangles preprints. The dashed line is the pre-specified interpretation threshold (2 pp).

Table 4: The same six methods on MedMCQA, measured against two different comparators [Authors, 2026]. The sign of every conclusion flips.

Method (pool of 7–13B experts)	Acc. (%)	vs 7B single	vs 72B single
Self-Refine	74.58	+2.51	−5.86
MoE router	75.45	+3.38	−4.99
MAD (debate)	76.55	+4.48	−3.89
GraphRouter	76.92	+4.85	−3.52
S-DAG (proposed)	78.38	+6.31	−2.06
Symbolic-MoE	78.55	+6.48	−1.89
Median		+4.67	−3.71

support for the hypothesis that smaller models benefit more; the evidence is too sparse and too confounded by fine-tuning to decide.

3.4 RQ3: what is bought, and at what price

Figure 3 plots ΔAcc against the derived cost ratio. Cost could be established for 110 of 293 comparisons, and in only 28 of those was it *reported* by the primary study rather than derived from the architecture description. Typical multi-agent configurations sit at a median R of $4\times$ (110 comparisons with a numeric cost ratio). The highest cost-efficiency values are obtained by cheap mechanisms: Tang et al. [2023] against a self-consistency baseline ($\Delta = +9.2$ pp at $R \approx 1.4$, $\text{CEI} \approx 19$) and the two-stage self-correction of Su et al. [2025] ($\Delta = +18.5$ pp at $R = 2$, $\text{CEI} \approx 18$). Large orchestrations buy proportionally less: at $R = 20$ the gain on hard items is $+9.0$ pp, i.e. $\text{CEI} \approx 2$.

Two convergent data points concern aggregation. In Chen et al. [2025] the multi-agent system reaches 90.1% on MedQA while *Medprompt*, a single-agent prompting strategy, reaches 90.2% on the same slice — -0.1 pp for a substantially more complex system. In a peer-reviewed licensing-examination study [pm4, 2025], an adaptive multi-agent system reached 89.97% while chain-of-thought plus few-shot prompting reached 87.67%; the authors themselves describe the

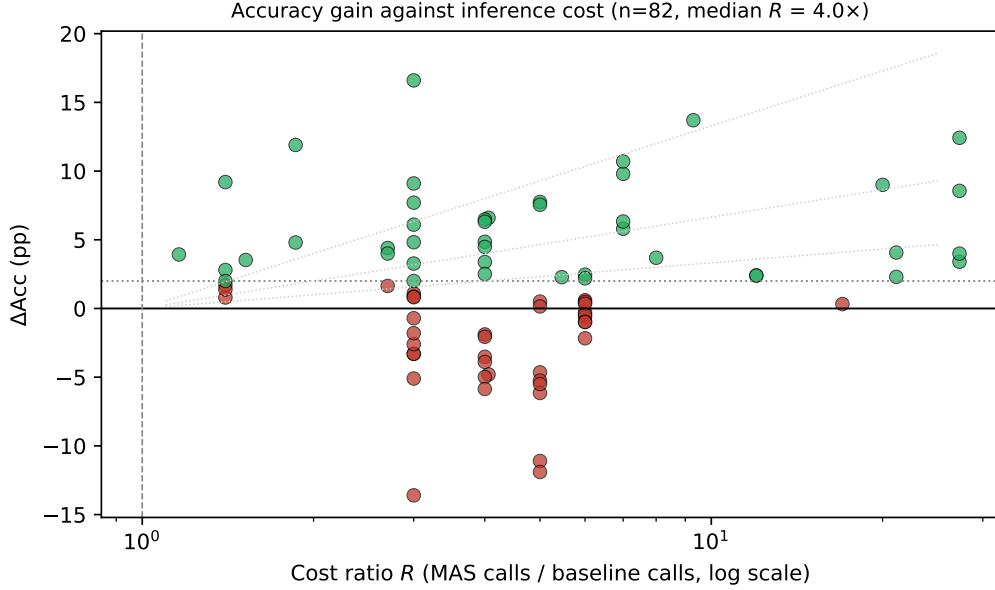


Figure 3: Accuracy gain against derived cost ratio (log scale). Green points exceed the 2 pp threshold, red points do not. Dotted lines are iso-efficiency contours.

latter as offering “an optimal balance between performance and efficiency” because it achieves this “with minimal API usage”. A peer-reviewed agentic benchmark in *npj Digital Medicine* [pm4, 2026] reports that adding agent enhancements moved accuracy from 24.0% to 28.0% on one clinical task *without statistical significance* — a published null result of exactly the kind that the preprint literature rarely contains.

3.5 RQ4: gains concentrate on hard items

Two studies stratify gains by item difficulty, and they agree.

The medical evidence comes from MedLA [Ma et al., 2026], which evaluates four published multi-agent methods and a same-backbone LLaMA-3.1-8B single agent on the MedDDx benchmark at three difficulty levels (Figure 4). Excluding the proposing system, the median gain is -3.1 pp on *Basic* items, -2.2 pp on *Intermediate* items and $+1.4$ pp on *Expert* items. In other words, on easy and moderate medical questions multi-agent collaboration is actively *harmful* relative to simply querying the same model once, and only at expert difficulty does it turn positive — and even then by less than the 2 pp threshold. Majority voting is negative at all three levels.

The general-domain evidence agrees in direction and is larger in magnitude: scaling test-time compute $15\text{--}20\times$ buys $+2.2$ pp on easy items, $+8.5$ pp on medium items and $+9.0$ pp on hard items [Wunderlich et al., 2026]; on easy items essentially nothing is bought for a twentyfold cost. The same study reports that mixture-of-agents reaches 71.4% versus 64.3% for chain-of-thought, while self-consistency reaches 68.7% and debate 70.0% under comparable compute — again placing a non-interactive control within about one point of the interactive methods.

3.6 Reporting practice

Cost reporting remains the binding constraint: across 293 extracted comparisons a numeric cost ratio exists for 110, and only 28 of those come from a figure the primary study reported (API calls, tokens, USD or expert invocations); the remaining 82 were derived from the architecture description, and 177 comparisons carry no usable cost information at all. Sampling procedure — whether the full test split or a subsample was used, and with what seed — was frequently

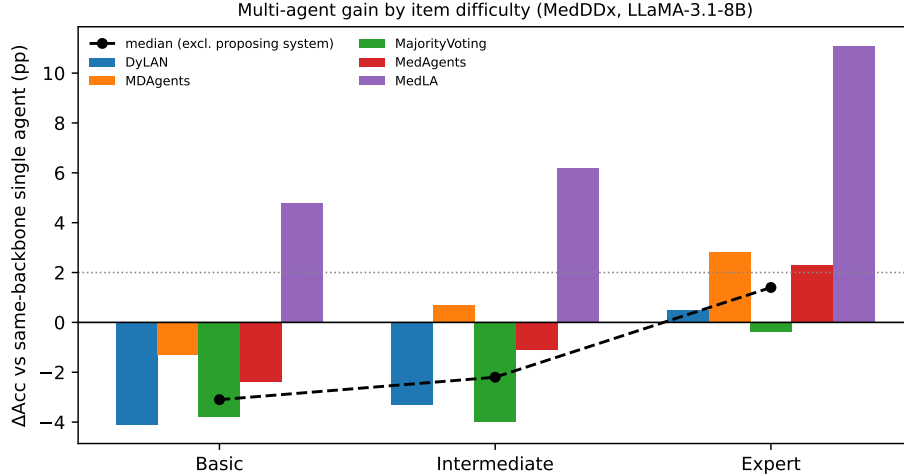


Figure 4: Multi-agent gain against a same-backbone single agent by item difficulty (MedDDx, LLaMA-3.1-8B). The dashed line is the median across methods excluding the proposing system: negative on Basic and Intermediate items, marginally positive on Expert items. Data from Ma et al. [2026].

unstated, which prevents several otherwise comparable results from being placed in the same evaluation group.

4 Discussion

4.1 Principal findings

Three findings survive the analysis. (i) The headline accuracy gains reported for medical multi-agent systems do not survive a fair comparator: measured against the strongest single agent the same paper already ran, the study-level median is -0.31 pp with a 95% CI of $[-2.70, +0.36]$, i.e. no reproducible gain in either direction, and 69% of individual comparisons are negative. (ii) Non-interactive resampling (self-consistency) is competitive with, and sometimes superior to, interactive debate and mixing at comparable cost — which means that a substantial part of the reported “collaboration” benefit is purchased compute rather than collaboration. (iii) What benefit exists is confined to hard items: on easy and moderate medical questions the median effect against a same-backbone single agent is *negative* (-3.1 and -2.2 pp), turning marginally positive ($+1.4$ pp) only at expert difficulty.

Taken together these support a simple practical rule: *spend compute selectively on hard items, prefer cheap aggregation, and always report what the strongest single agent would have achieved.*

4.2 Limitations of the evidence

The included studies are now overwhelmingly peer-reviewed (280 of 293 extracted comparisons), so the pattern cannot be attributed to preprint quality. It is instead attributable to *who chooses the comparator*: in almost every study the baseline is selected by the same authors who propose the method, which is precisely the mechanism that produces finding (i). The two studies that were explicitly designed as neutral benchmarks rather than as method papers — MedicalAgentsBench and MedAgentBoard — are also the two that report the most negative results. Cost is rarely measured. Variance is almost never reported, so no pooled estimate with an interval is possible. Item-level results are almost never released, which is why RQ4 rests on a single study and on a general-domain benchmark rather than a medical one. Benchmark contamination is checked in a minority of studies.

4.3 Limitations of the review process

This draft deviates from its own protocol in ways that must be stated plainly. The protocol was written but **not prospectively registered**. Screening, data extraction and appraisal were performed by a **single reviewer** without duplicate checking, and the formal risk-of-bias assessment specified in the protocol (six domains) has **not** been completed.

One study dominates. MedicalAgentsBench contributes 36 of the 93 primary comparisons; removing it moves the comparison-level median from -1.73 to -0.60 pp. The sign is stable across all thirteen leave-one-study-out refits, the magnitude is not.

Difficulty and benchmark are confounded. A large share of the comparisons come from deliberately hard-curated subsets on which every method operates near the floor, where paired differences are noisier and systematically unlike full-test-split behaviour.

Variances are essentially never recoverable, so no pooled effect with a conventional interval is possible; the bootstrap reported here propagates between-comparison and between-study variation only, not within-comparison sampling error.

The screened corpus came from two sources, not three. 1,129 OpenAlex records were retrieved but never entered the pre-specified triage rule because the API returns titles without retrievable abstracts. A post hoc title-level screen of the 1,070 unscreened records passed 0 on a three-facet screen and 3 on a two-facet screen, all clearly ineligible, so the residual risk is small but the search sensitivity is bounded below rather than established. The main search also required a supplementary round of probes to recover well-known systems that the primary facet queries had missed.

Retrieval remains incomplete. Fifty subscription articles could not be retrieved, and for a further 67 peer-reviewed articles only full text, not the publisher PDF, could be obtained; both groups are recorded in the register for manual retrieval.

Extraction is a convenience subset. Of the studies for which a medical benchmark, a baseline and extractable accuracies could all be identified, 30 contributed the 293 rows analysed here; the remainder are registered and await extraction.

Accordingly this document should be read as a **preliminary synthesis** that fixes the analytical frame and reports the signal visible in the extracted subset, not as a completed systematic review.

Accordingly, this document should be read as a **preliminary synthesis** that fixes the analytical frame and reports the signal visible in the first extracted subset, not as a completed systematic review.

4.4 Implications

For method developers: report (a) the strongest single-agent baseline using the same backbone, same evaluation items and same prompting budget; (b) an equal-budget self-consistency control arm; (c) per-question calls, tokens and, where applicable, cost; (d) item-level predictions. For reviewers and editors: a multi-agent result without a cost figure and without a same-backbone strongest baseline is not interpretable. For future reviews: the quantities that make synthesis possible — paired item-level outcomes and cost — are precisely those that are missing, and that gap is itself the most robust finding available today.

5 Conclusion

On the evidence retrievable today, the accuracy advantage of LLM multi-agent systems in medical examination question answering is smaller than the literature suggests, is concentrated on difficult items, and is frequently obtainable more cheaply by non-interactive resampling. The field does

not yet report enough cost information to identify a balance point; making that reporting routine is the precondition for any meaningful recommendation about which architecture to deploy.

References

- Performance of single-agent and multi-agent language models in Spanish language medical competency exams. *BMC medical education*, 2025. doi: 10.1186/s12982-025-00429-5. PMID:40336004.
- Benchmarking large language model-based agent systems for clinical decision tasks. *NPJ digital medicine*, 2026. doi: 10.1038/s41597-022-01899-x. PMID:41708802.
- S-DAG Authors. S-DAG: A Subject-Based Directed Acyclic Graph for Multi-Agent Heterogeneous Reasoning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. arXiv:2511.06727.
- Kai Chen, Xinfeng Li, Tianpei Yang, Hewei Wang, Wei Dong, and Yang Gao. MDTeamGPT: A Self-Evolving LLM-based Multi-Agent Framework for Multi-Disciplinary Team Medical Consultation, 2025.
- Shuo Ma, Jia Huang, Fan Zhang, Jun Wu, Yi Shen, and Guodong Fan. MedLA: A Logic-Driven Multi-Agent Framework for Complex Medical Reasoning with Large Language Models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. arXiv:2509.23725.
- Niloufar Mehrabi, Sayed Pedram Haeri Boroujeni, and Abolfazl Razi. From Talking Words to Sharing Thoughts: Scalable Multi-LLM Aggregation via Structured Message Passing, 2026.
- Xinchun Su, Chunxu Luo, Yixuan Li, Weidong Yang, and Lipeng Ma. MedCritical: Enhancing Medical Reasoning in Small Language Models via Self-Collaborative Correction, 2025.
- Xiangru Tang, Anni Zou, Zhuosheng Zhang, Ziming Li, Yilun Zhao, Xingyao Zhang, Arman Cohan, and Mark Gerstein. MedAgents: Large Language Models as Collaborators for Zero-shot Medical Reasoning. In *Annual Meeting of the Association for Computational Linguistics*, 2023. arXiv:2311.10537.
- Xiangru Tang, Daniel Shao, Jiwoong Sohn, Jiapeng Chen, Jiayi Zhang, Jinyu Xiang, Fang Wu, Yilun Zhao, Chenglin Wu, Wenqi Shi, Arman Cohan, and Mark Gerstein. MedicalAgentsBench for complex medical reasoning: Comparing internalized reasoning models versus externalized agent-based frameworks. *Patterns*, 7:101658, 2026. doi: 10.1016/j.patter.2026.101658.
- Florian Valentin Wunderlich, Lars Benedikt Kaesberg, Jan Philip Wahle, Terry Ruas, and Bela Gipp. Multi-Agent Reasoning Improves Compute Efficiency: Pareto-Optimal Test-Time Scaling. In *Annual Meeting of the Association for Computational Linguistics*, 2026. arXiv:2605.01566.
- Sukwon Yun, Jie Peng, Pingzhi Li, Wendong Fan, Jie Chen, James Zou, Guohao Li, and Tianlong Chen. Graph-of-Agents: A Graph-based Framework for Multi-Agent LLM Collaboration. In *International Conference on Learning Representations*, 2026. arXiv:2604.17148.
- Yinghao Zhu, Ziyi He, Haoran Hu, Xiaochen Zheng, Xichen Zhang, Zixiang Wang, Junyi Gao, Liantao Ma, and Lequan Yu. MedAgentBoard: Benchmarking Multi-Agent Collaboration with Conventional Methods for Diverse Medical Tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. arXiv:2505.12371.

A Data availability

Search outputs (`search/`), triage and screening records (`screening/`), the extracted comparison table (`extraction/data.csv`), analysis scripts (`analysis/`) and the document register (`documents.md`) accompany this draft.